

Selection control structures

1. What is a control structure in programming?
 - a. To define the flow of execution in a program
 - b. To store data in memory
 - c. To represent graphical output
 - d. To connect external devices
2. What is the purpose of conditional statements in a program?
 - a. To repeat a block of code
 - b. To execute code based on a condition
 - c. To store large amounts of data
 - d. To display graphical elements
3. In an "if" statement, when does the code block execute?
 - a. When the condition is false
 - b. When the condition is true
 - c. Regardless of the condition
 - d. Only when a loop is running
4. How is an "if-else" statement different from an "if" statement?
 - a. It doesn't allow conditions
 - b. It has an additional block for false conditions
 - c. It runs all blocks of code simultaneously
 - d. It can only check numerical values
5. What is a nested "if" statement?
 - a. An "if" statement with no condition
 - b. An "if" statement inside another "if" statement
 - c. A condition without an "else" block
 - d. A switch-case inside an "if" statement
6. In which situation would you use a nested "if" statement?
 - a. When checking multiple conditions that depend on each other
 - b. When a single condition is sufficient
 - c. When a loop is required
 - d. When a variable is undefined
7. What is the purpose of the "else" keyword?
 - a. To define a condition
 - b. To repeat the code block
 - c. To provide an alternative action when a condition is false
 - d. To initialize variables

8. How many "else" blocks can an "if" statement have?
 - a. None
 - b. One
 - c. Two
 - d. Multiple
9. Can an "if-else" structure have multiple "else" blocks?
 - a. Yes
 - b. No
 - c. It depends on the language
 - d. Only with nested "if" statements
10. What is a switch-case statement used for?
 - a. To store data
 - b. To evaluate multiple conditions based on a single variable
 - c. To repeat code
 - d. To define a function
11. How is a switch-case different from an "if-else" structure?
 - a. Switch-case is more efficient for checking a single variable with many possible values
 - b. Switch-case can only be used with numbers
 - c. Switch-case is more flexible than "if-else"
 - d. Switch-case does not need a condition
12. What is the purpose of the "break" statement in switch-case?
 - a. To break the switch-case structure and continue with the rest of the program
 - b. To exit a loop
 - c. To stop the program execution
 - d. To terminate a function
13. What happens if the "break" statement is omitted in a switch-case?
 - a. The code will skip all cases
 - b. The program will terminate
 - c. The cases will "fall through" and execute one after another
 - d. The program will crash
14. What is the purpose of the "default" case in a switch-case structure?
 - a. To handle cases that don't match any of the specified "case" values
 - b. To terminate the program
 - c. To store the program's output
 - d. To repeat the case execution

15. What type of data is typically used in switch-case conditions?
- a. Strings
 - b. Booleans
 - c. Integers or characters
 - d. Arrays
16. Which type of structure is used to handle a menu-driven program?
- a. Switch-case
 - b. For loop
 - c. While loop
 - d. Recursion
17. What will be the output of the following code?
- ```
int x = 5;
if (x > 3)
if (x < 10)
 printf("A");
else
 printf("B");
```
- a. A
  - b. B
  - c. A followed by B
  - d. No output

18. What will be the output of the following code?

```
int x = 2;

switch (x) {

 case 1: printf("One");

 case 2: printf("Two");

 case 3: printf("Three");

 default: printf("Default");

}
```

- a. Two
- b. Two Three Default
- c. Default
- d. Compilation error

Explanation: The switch statement does not break after each case, so it executes all cases from the matched one (case 2) to the default.

19. Which of the following scenarios is best handled by a nested if-else structure?

- a. Checking if a number is even or odd.
- b. Validating a password and checking if it has special characters.
- c. Iterating over a list of numbers.
- d. Printing the first 10 multiples of a number.

Explanation: A nested if-else is suitable when multiple related conditions must be checked sequentially.

20. What will the following code output?

```
int x = 4;
switch (x) {
 case 1: printf("One"); break;
 case 2: printf("Two"); break;
 case 3: printf("Three"); break;
 default: printf("None");
}
```

- a. One
- b. Two
- c. Three
- d. None

Explanation: Since there is no case for  $x = 4$ , the control goes to the default block, and "None" is printed.